

Cabibbo-Kobayashi-Maskawa Matrix

$$\begin{pmatrix} u \\ d' \end{pmatrix} \quad \begin{pmatrix} c \\ s' \end{pmatrix}$$

$$d' = d \cdot \cos \theta + s \cdot \sin \theta$$

$$s' = -\sin \theta \cdot d + s \cdot \cos \theta$$

$$u \quad d \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} c \\ s \end{pmatrix}$$

1 real parameter: Cabibbo angle

$$\sin \theta \approx 0.25$$

u $\xrightarrow{g \cos \theta}$ d
 $\Delta S = 0$
 Cabibbo-Favored

u $\xrightarrow{g \sin \theta}$ s
 $\Delta S = 1$
 Cabibbo-suppressed

c $\xrightarrow{-g \sin \theta}$ d
 Suppressed
 $\Delta C = 1$
 $\Delta S = 0$

c $\xrightarrow{g \cos \theta}$ s
 Favored.
 $\Delta C = 1$
 $\Delta S = 1$

$$\begin{pmatrix} u \\ d \end{pmatrix} \quad \begin{pmatrix} c \\ s \end{pmatrix}$$

transitions within same family favored $\propto \cos \theta$
 change of family suppressed $\propto \sin \theta$

1964: Cronin-Fitch experiment
 CP Violation in Kaons.

$$K^0 \rightarrow \pi^+ \pi^-$$

$$K^0 \rightarrow \pi^+ \pi^- \pi^0$$

~~CP~~

1973: Kobayashi-Maskawa proposed existence of a third family of quarks.

— no charm observed yet (until late 1974)

$$\begin{pmatrix} u \\ d \end{pmatrix} \quad \begin{pmatrix} c \\ s \end{pmatrix} \quad \begin{pmatrix} t \\ b \end{pmatrix}$$

c: need for $K^0 \rightarrow \pi^+ \pi^-$ suppression.

b.t. needed to explain CP in Kaons.

1977: b quark discovery

1994: top discovery.

$$q_i: \frac{g_w V_{ij}}{\sum w^*} q_j \quad g_w \bar{\psi}_i V_{ij} \gamma^\mu (1-\gamma^5) \psi_j$$

$$V_{ij} \equiv V_{CKM}$$

V_{CKM} : rotation matrix for fields of quarks.

properties: V_{CKM} must be unitary.

$$\begin{pmatrix} u & c & t \end{pmatrix} \begin{pmatrix} V_{ud} = \cos\theta & V_{us} = \sin\theta & V_{ub} \\ V_{cd} = -\sin\theta & V_{cs} = \cos\theta & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix} \begin{pmatrix} d \\ s \\ b \end{pmatrix}$$

$$V_{CKM}^\dagger V_{CKM} = \mathbb{1}$$

3x3 complex matrix $\Rightarrow$ 18 total parameters

$$V^\dagger V = \mathbb{1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \Rightarrow \begin{matrix} 3 \text{ diagonals} = 1 \\ 6 \text{ conditions} \\ \text{off diagonal.} \end{matrix}$$

$$V_{CKM} = V_{\text{rot}}^{\text{up-like}} \cdot V_{\text{rot}}^{\text{down-like}}$$

$$q_j \rightarrow e^{i\alpha_j} q_j \quad q_j: \text{field of quark } j$$

$\Rightarrow$ arbitrary choice of 5 phases.
only 1 phase remains.

$$\Rightarrow 9 - 5 = 4 \text{ parameters.}$$

$$\Rightarrow 3 \text{ real parameters / Euler angles.}$$

$$1 \text{ complex phase.}$$

4 complex phase needed to explain $\bar{CP}$ in kaons.

Why 3 families needed?

Suppose only 2 families:

$$\begin{pmatrix} u \\ d \end{pmatrix} \begin{pmatrix} c \\ s \end{pmatrix}$$

$$\overline{u c} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} d \\ s \end{pmatrix}$$

2x2 complex matrix $\Rightarrow$ 8 parameters.

unitarity cond $\Rightarrow$ 2 diagonal = 1

2 off-diag. = 0

$\Rightarrow$ 4 left.

4 quark fields $\Rightarrow$ 3 can be chosen arbitrarily.
1 phase remnant.

$\Rightarrow$ 4 - 3 = 1 param.

Cabibbo angle.

NxN matrix.

2N-1 param.

can be
fixed
orbit

$$\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

$\Rightarrow$ no complex phase in matrix.

Kobayashi-Maskawa: complex phase needed
to explain CP in kaons.

$$\begin{aligned} V_{CKM} &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix} \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta} & 0 & c_{13} \end{pmatrix} \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix} \end{aligned}$$

$$V_{\text{CKM}} = \begin{pmatrix} 1 - \lambda^2/2 & \lambda & A\lambda^3(\rho - i\eta) \\ -\lambda & 1 - \lambda^2/2 & A\lambda^2 \\ A\lambda^3(1 - \rho - i\eta) & -A\lambda^2 & 1 \end{pmatrix} + \mathcal{O}(\lambda^4)$$

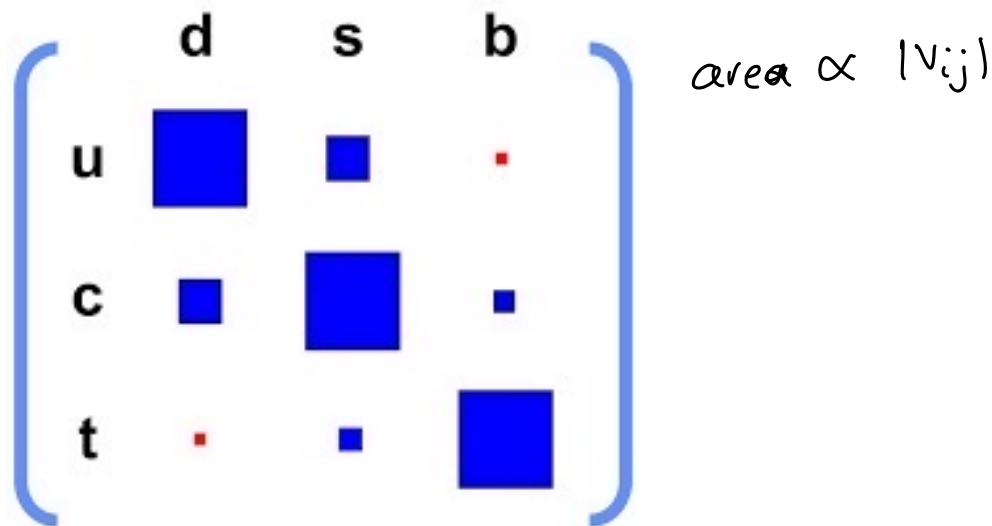
wolfenstein parameterization.

$$\lambda = \sin\theta_c \approx 0.25$$

$$\sin^2\theta_c = 0.04$$

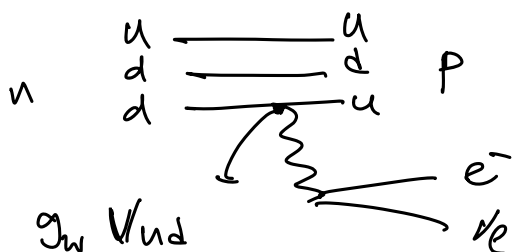
$$V_{\text{CKM}} \equiv V_L^u V_L^{d\dagger} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix}$$

Charged weak current? favors transitions in same family



$$|V_{\text{CKM}}| = \begin{pmatrix} 0.97435 \pm 0.00016 & 0.22500 \pm 0.00067 & 0.00369 \pm 0.00011 \\ 0.22486 \pm 0.00067 & 0.97349 \pm 0.00016 & 0.04182^{+0.00085}_{-0.00074} \\ 0.00857^{+0.00020}_{-0.00018} & 0.04110^{+0.00083}_{-0.00072} & 0.999118^{+0.000031}_{-0.000036} \end{pmatrix}$$

Measurement of V_{CKM} elements.



$$\Gamma(n \rightarrow p e^- \bar{\nu}_e) \propto |M|^2 \sim |V_{ud}|^2$$

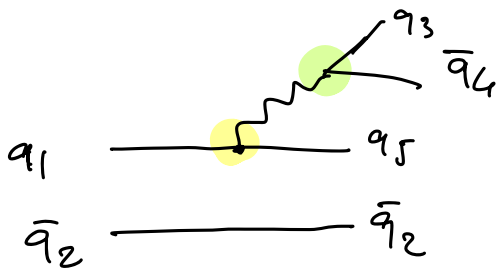
$$\hookrightarrow \text{BF}(n \rightarrow p e^- \bar{\nu}_e) \cdot \frac{1}{\tau} \text{ is measured}$$

$$\Rightarrow \text{compute } |V_{ud}|$$

$$\begin{pmatrix}
 \mathbf{V}_{ud} & \mathbf{V}_{us} & \mathbf{V}_{ub} \\
 \pi \rightarrow \ell \nu & K \rightarrow \ell \nu & B \rightarrow \pi \ell \nu \\
 & K \rightarrow \pi \ell \nu & \\
 \mathbf{V}_{cd} & \mathbf{V}_{cs} & \mathbf{V}_{cb} \\
 D \rightarrow \ell \nu & D_s \rightarrow \ell \nu & B \rightarrow D \ell \nu \\
 D \rightarrow \pi \ell \nu & D \rightarrow K \ell \nu & B \rightarrow D^* \ell \nu \\
 \mathbf{V}_{td} & \mathbf{V}_{ts} & \mathbf{V}_{tb} \\
 B_d \leftrightarrow \bar{B}_d & B_s \leftrightarrow \bar{B}_s &
 \end{pmatrix}$$

to measure $V_{ij} \Rightarrow$ need hadrons with q_i, q_j

Mesons preferred over baryons: smaller QCD corrections.
Semileptonic decays preferred to hadronic decays.



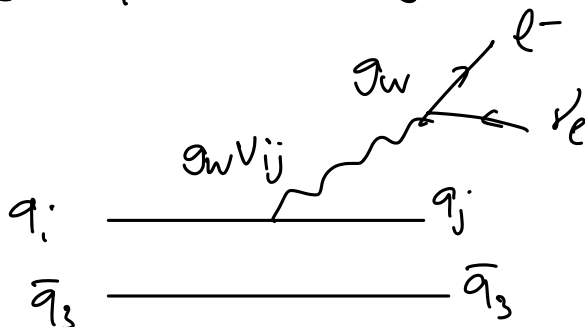
hadronic decay.

$$\mathcal{M} \propto V_{15} V_{34}$$

$$\Gamma \propto |V_{15}|^2 |V_{34}|^2$$

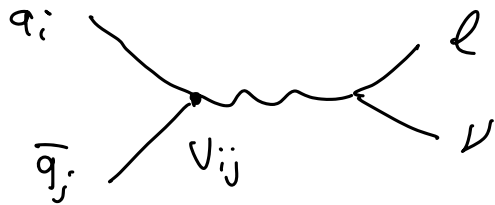
measure Γ quantity
experimentally

In Semileptonic decays: 1 CKM element



$$\Gamma \propto |V_{ij}|^2$$

leptonic decays.

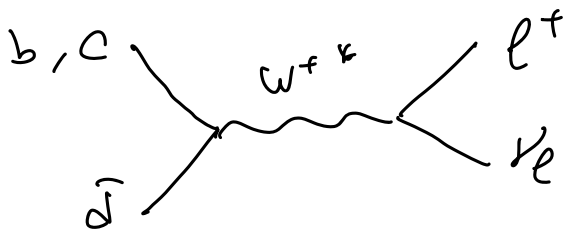


$$\Gamma \propto |V_{ij}|^2$$

Usually $\Gamma(P \rightarrow e\nu) \ll \Gamma(P \rightarrow P' e\nu)$
 P, P' : mesons.

$$\pi^+ \rightarrow \mu\nu \quad \sim 100\%$$

$$K^+ \rightarrow \mu\nu \quad \sim 64\%$$



$$D^+ \rightarrow e\nu$$

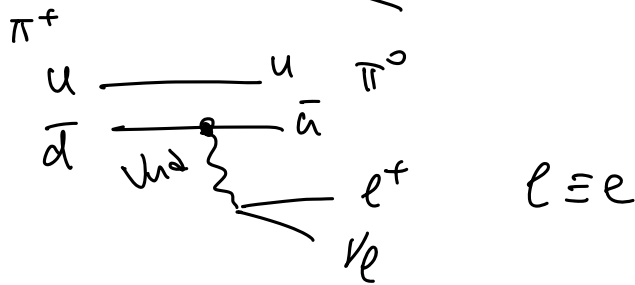
$$B^+ \rightarrow e\nu$$

} $B\Gamma$ = very small.

V_{ud}



$$\Gamma \propto |V_{ud}|^2 \times (\text{QCD correction})$$



$$\pi^+ \rightarrow \pi^0 e^+ \nu_e$$

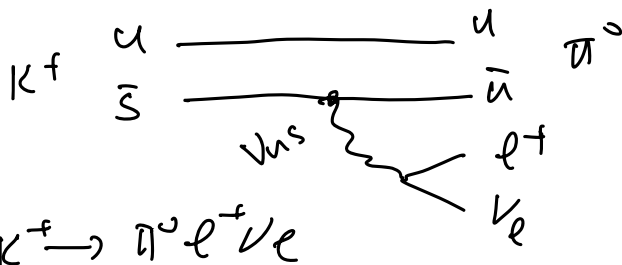
$$\Gamma \propto |V_{ud}|^2$$



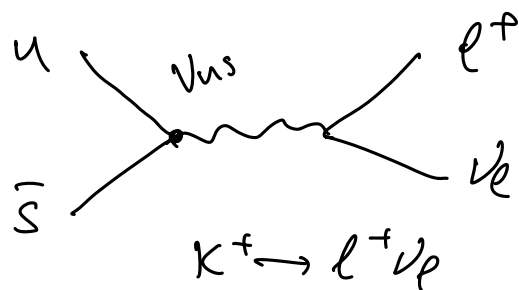
$$\pi^+ \rightarrow \mu^+ \nu_\mu$$

$$\Gamma \propto |V_{ud}|^2$$

V_{us}

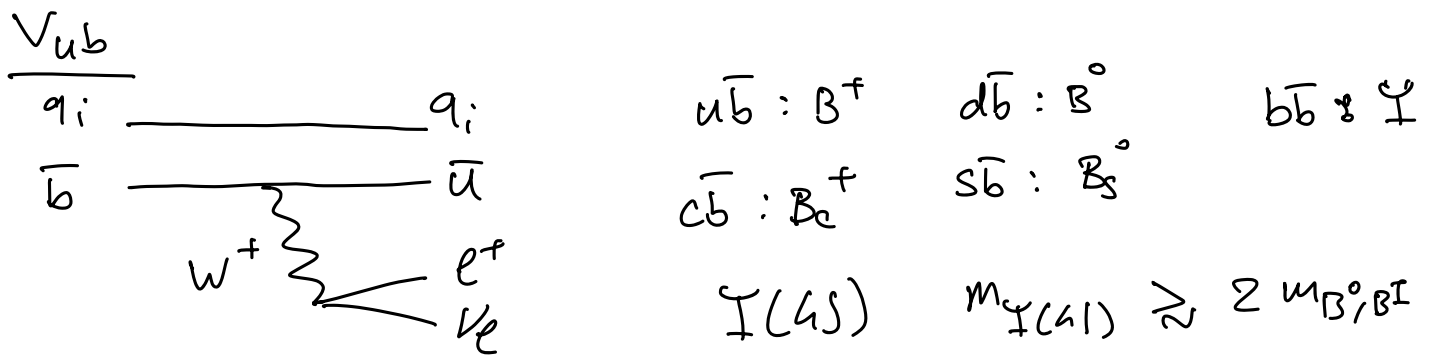


$$K^+ \rightarrow \pi^0 e^+ \nu_e$$



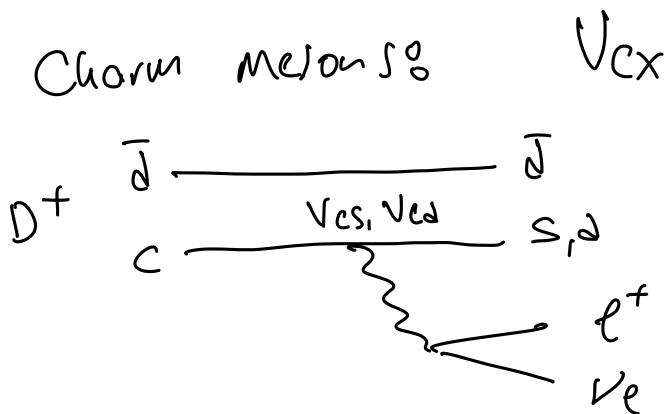
$$K^+ \rightarrow e^+ \nu_e$$

K^+ DECAY MODES	Fraction (Γ_i/Γ)	Scale factor/ Confidence level (MeV/c)	p
Leptonic and semileptonic modes			
$e^+ \nu_e$	$(1.582 \pm 0.007) \times 10^{-5}$		247
$\mu^+ \nu_\mu$	$(63.56 \pm 0.11) \%$	$S=1.2$	236
$\pi^0 e^+ \nu_e$	$(5.07 \pm 0.04) \%$	$S=2.1$	228
Called K_{e3}^+ .			
$\pi^0 \mu^+ \nu_\mu$	$(3.352 \pm 0.033) \%$	$S=1.9$	215
Called $K_{\mu 3}^+$.			
$\pi^0 \pi^0 e^+ \nu_e$	$(2.55 \pm 0.04) \times 10^{-5}$	$S=1.1$	206
$\pi^+ \pi^- e^+ \nu_e$	$(4.247 \pm 0.024) \times 10^{-5}$		203
$\pi^+ \pi^- \mu^+ \nu_\mu$	$(1.4 \pm 0.9) \times 10^{-5}$		151
$\pi^0 \pi^0 \pi^0 e^+ \nu_e$	$< 3.5 \times 10^{-6}$	$CL=90\%$	135
Hadronic modes			
$\pi^+ \pi^0$	$(20.67 \pm 0.08) \%$	$S=1.2$	205
$\pi^+ \pi^0 \pi^0$	$(1.760 \pm 0.023) \%$	$S=1.1$	133
$\pi^+ \pi^+ \pi^-$	$(5.583 \pm 0.024) \%$		125



$$e^+ e^- \rightarrow \Upsilon(4S) \rightarrow \begin{array}{l} B^0 \bar{B}^0 \\ B^+ B^- \end{array}$$

$$\Gamma(B \rightarrow \pi \ell \nu) \propto |V_{ub}|^2$$



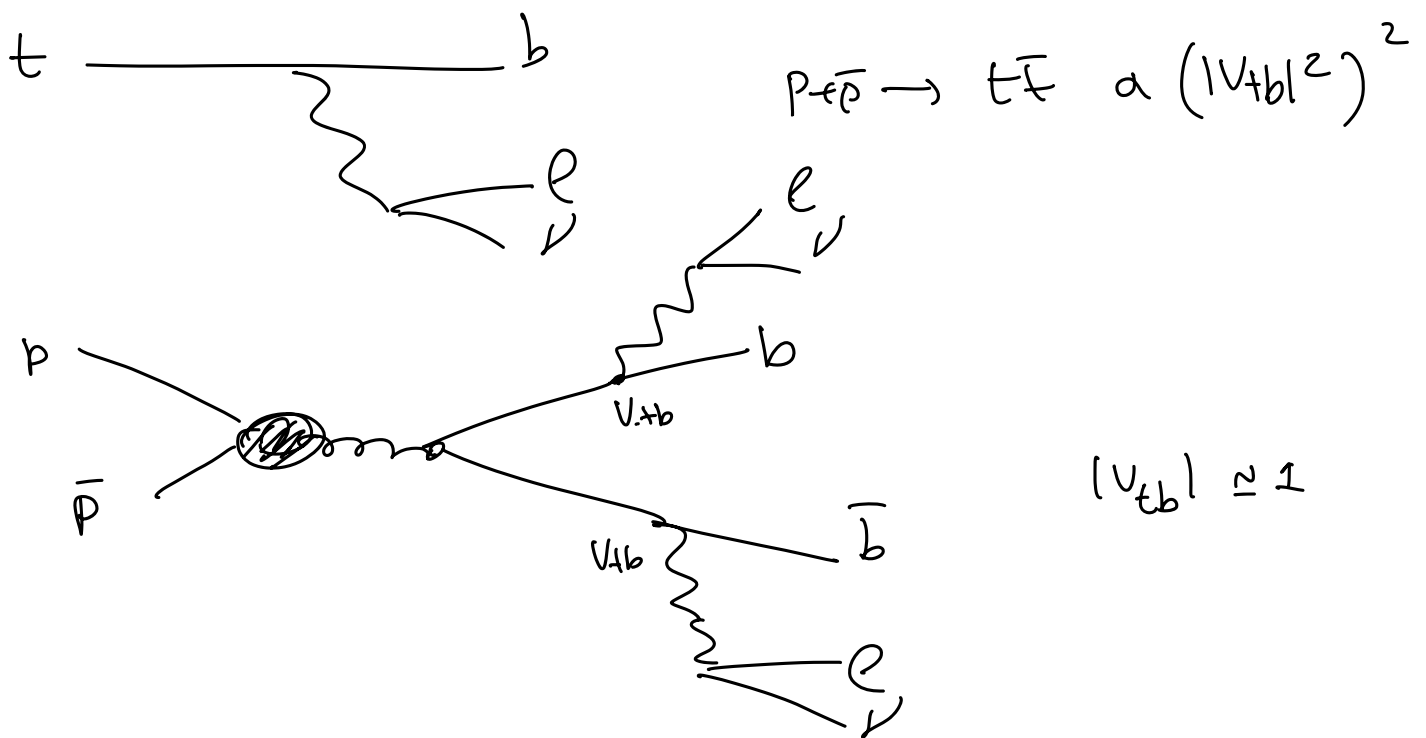
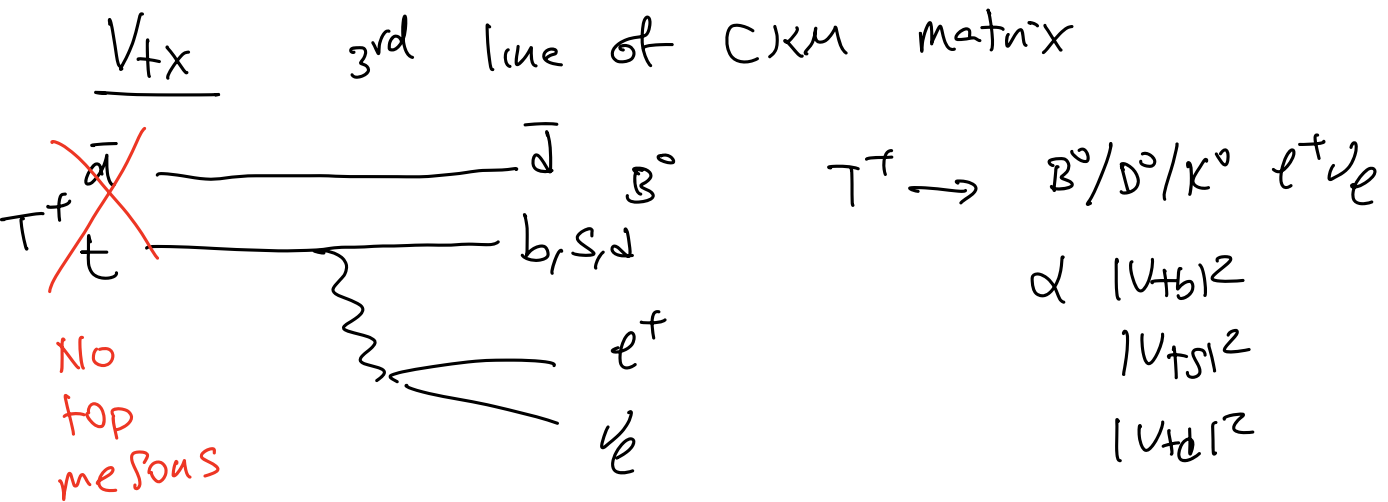
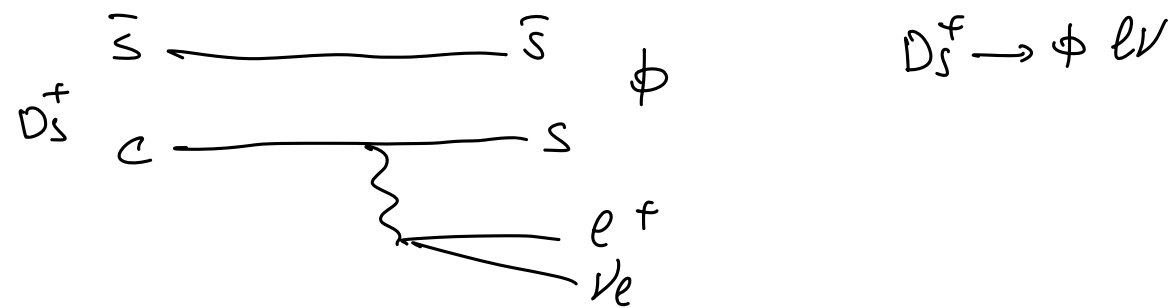
$$c\bar{d} : D^+$$

$$c\bar{s} : D_s^+$$

$$u\bar{c} : D^0$$

$$D^+ \rightarrow \bar{K}^0 \ell \nu \propto |V_{cs}|^2$$

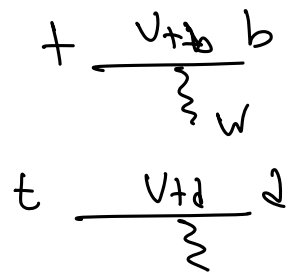
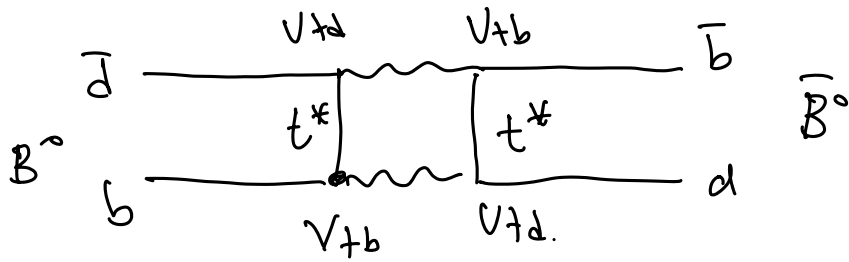
$$\pi^0 \ell \nu \propto |V_{cd}|^2$$



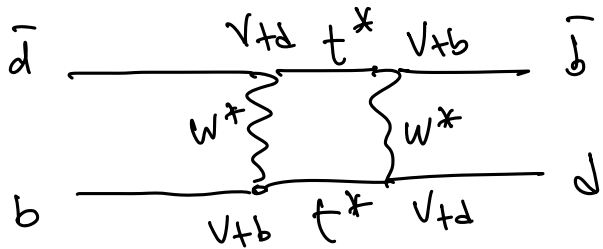
At leading order (tree level) no transition to measure V_{tx} .

$\Rightarrow$ Look at higher order processes.

Use B mesons to measure V_{tb}



Box diagram.



$$\mathcal{M} \propto |V_{tb}|^2 |V_{td}|^2$$

$$\Delta b = 2$$

$B^0 \leftrightarrow \bar{B}^0$ oscillation.

Same for s, c quarks

$$D^0 \leftrightarrow \bar{D}^0$$

$$K^0 \leftrightarrow \bar{K}^0$$

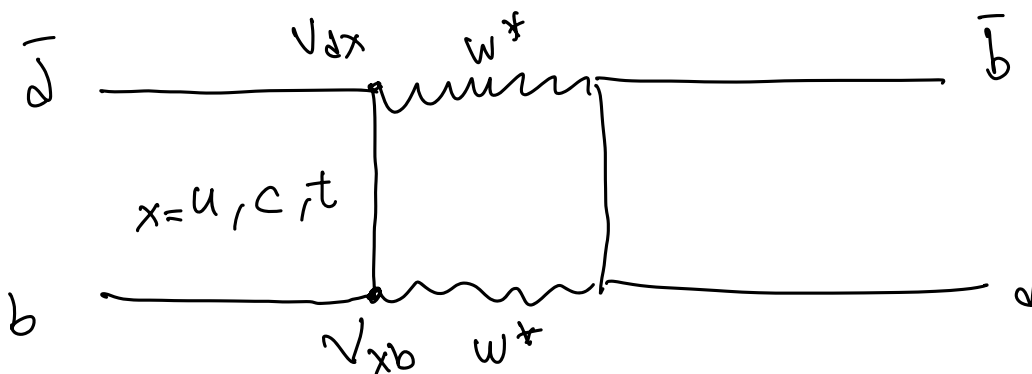
$$m_{B^0} \equiv m_{\bar{B}^0}$$

$$m_{D^0} \equiv m_{\bar{D}^0}$$

$$m_{K^0} \equiv m_{\bar{K}^0}$$

multiple measurements needed to extract $|V_{tb}|, |V_{td}|, |V_{ts}|$

Decay rates sensitive only to $|V_{ij}|$



top quark dominates the box diagram

$$\begin{pmatrix} u \\ d \end{pmatrix} \begin{pmatrix} c \\ s \end{pmatrix} \begin{pmatrix} t \\ b \end{pmatrix}$$